Chapter 2. Populations and Samples

Distributions, summary measures and graphical displays

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2.1 Distribution of a population

Motivation. Before estimating anything we need a language for "what the whole population looks like". The distribution records how the characteristic values are spread, as nonnegative weights summing to one.

Table 1-1 summarized a *sample*; the actual election result summarizes the *population*:

Candidate	J. Smith	E. Johnson	M. Lee	S. Davis	Total
Vote share	0.367	0.281	0.271	0.081	1.000

The shares are nonnegative and sum to 1; drawn as bars they form a bar graph (Figure 2-1).

Definition 2-1 (Distribution of a population)

The way the characteristic values of a population are spread, expressed by nonnegative numbers summing to 1, is called the **distribution** of the population.

A main purpose of a pre-election poll is to estimate this distribution.

Illustration 2-1: Mendel's peas

Let R (round), r (wrinkled), Y (yellow), y (green) be pea genes. Crossing RRYY with rryy, the second generation shows round–yellow (C1), round–green (C2), wrinkled–yellow (C3), wrinkled–green (C4) in ratio $9:3:3:1$. Repeating the crossing indefinitely gives an infinite population with values C1–C4, whose distribution (if Mendel is right) is

Value	C1	C2	C3	C4
Proportion	9/16	3/16	3/16	1/16

Qualitative values (candidate, pea shape and colour) or discrete values (number of children of married women) have a distribution given by relative frequencies, or their limits for infinite populations.

Continuous characteristics. If the characteristic is height, weight, etc., which can take any value in an interval, group the values into classes. With class width 10 cm suppose the relative frequencies are

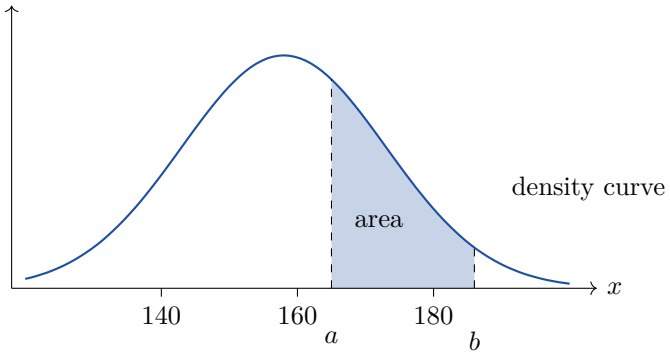
Class (cm)	130–140	140–150	150–160	160–170	170–180	180–190	Total
Rel. freq.	0.05	0.1	0.3	0.3	0.2	0.05	1.00

(class $[a, b)$, i.e. "a or more, less than b"). Assuming values are spread uniformly in each class, draw over each class a rectangle whose *area* is the class relative frequency (height = rel. freq./width); total area is 1. With width 5 cm the picture becomes finer (Table 2-4: relative frequencies $0.01, 0.04, 0.04, 0.06, 0.2, 0.1, 0.2, 0.1, 0.05, 0.15, 0.03, 0.02$ for $130, 135, \ldots, 185$), and its shape depends on the class choice. To remove this arbitrariness one describes the distribution by one smooth curve.

Definition (Density curve)

A curve describing the distribution of an infinite population with a continuous characteristic is a **density curve**. The area under it over $[a, b]$ equals the limit of the relative frequency

of values in $[a, b]$, so the total area between curve and horizontal axis is always 1. Even for a finite population, a density curve is a convenient approximation when the population is large.



Area over $[a, b]$ = limit of the relative frequency of values in $[a, b]$ (Figure 2-5)

For two or more characteristics the same idea applies: the distribution spreads total mass 1 over multivariate values—bar heights summing to 1 in the discrete case, volume 1 under a surface in the continuous case. This is a **joint distribution**.

2.2 Summary measures of a population

Motivation. A whole distribution is hard to estimate, so we summarize it by a few numbers describing location, spread and (for two characteristics) association. These only make sense for numerical characteristics.

2.2.1 Location: population mean, percentiles

For a finite population $\{c_1, \ldots, c_N\}$ (c_i the value of unit i), the **population mean** is

$$\mu = \frac{1}{N}(c_1 + c_2 + \cdots + c_N) = \frac{1}{N}\sum_{i=1}^{N} c_i.$$

If the distinct values are $c_1^*, \ldots, c_k^*$ with frequencies $f_1, \ldots, f_k$, then μ is the sum of (value)×(relative frequency):

Definition 2-2 (Population mean of a finite population)

$$\mu = \sum_{i=1}^{k} c_i^* \frac{f_i}{N}.$$

Example 2-1 (Number of children)

Number of children of the 16,549 thousand married Korean women aged 15+ (2020 census; 5* means "5 or more", counted as 5):

Children	0	1	2	3	4	5*
Rel. freq.	0.083	0.189	0.477	0.150	0.054	0.047

Find the population mean.

Solution.

$$\mu = 0(0.083) + 1(0.189) + 2(0.477) + 3(0.150) + 4(0.054) + 5(0.047) = 2.044,$$

about 2 children on average.

The mean is the **balance point**: placing at each value a mass proportional to its relative frequency, the bar graph balances at μ (Figure 2-7). The same interpretation holds for infinite populations: for a continuous characteristic, μ is the point where a plate shaped like the density curve balances.

Definition 2-3 (Population mean of an infinite population)

$$\mu = \begin{cases} \displaystyle\sum_{\text{all } x} x\,p(x) & \text{(discrete case)} \\ \displaystyle\int_{-\infty}^{\infty} x\,p(x)\,dx & \text{(continuous case)} \end{cases}$$

Here $p(x)$ is the limiting relative frequency of the value x (discrete case) or the density curve (continuous case); p is called the **density function**.

Example 2-2 (Mean of an infinite population)

Find the population mean of the infinite population

x	0	1	2	3	4
$p(x)$	1/10	3/10	3/10	2/10	1/10

Solution.

$$\mu = 0 \cdot \tfrac{1}{10} + 1 \cdot \tfrac{3}{10} + 2 \cdot \tfrac{3}{10} + 3 \cdot \tfrac{2}{10} + 4 \cdot \tfrac{1}{10} = 1.9.$$

The p**th percentile** is the point such that, with values in increasing order, $p\%$ are below it and $(100 - p)\%$ above. The 25th, 50th and 75th percentiles are the **first quartile** Q_1, the **median** Q_2 and the **third quartile** Q_3. Like μ, the median measures the centre: for a continuous infinite population it splits the area under the density curve in half. For a symmetric density $\mu = Q_2$; for a right-skewed density the long tail pulls μ to the right of Q_2 (Figure 2-9). Of these location measures, μ (together with σ below) is the key one; the median is usually not a target of inference because of limitations in estimation theory.

2.2.2 Spread: MAD, IQR, variance

For a finite population, the **mean absolute deviation** is the average distance from the median, and the **interquartile range** is the range of the middle 50%:

$$\text{MAD} = \frac{1}{N}\sum_{i=1}^{N} |c_i - Q_2|, \qquad \text{IQR} = Q_3 - Q_1.$$

Both are easy to interpret but are not usually inference targets. The standard measure is the **standard deviation** σ and its square, the **variance**:

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(c_i - \mu)^2}, \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(c_i - \mu)^2.$$

In terms of distinct values c_i^* with frequencies f_i:

Definition 2-4 (Variance and s.d. of a finite population)

(1) Population variance: $\sigma^2 = \sum_{i=1}^{k} (c_i^* - \mu)^2 \frac{f_i}{N}$.

(2) Population standard deviation: $\sigma = \sqrt{\sigma^2}$.

Definition 2-5 (Variance and s.d. of an infinite population)

For an infinite population with density function $p(x)$:

(1) Population variance:

$$\sigma^2 = \begin{cases} \sum_{\text{all } x} (x-\mu)^2 p(x) & \text{(discrete case)} \\ \int_{-\infty}^{\infty} (x-\mu)^2 p(x)\, dx & \text{(continuous case)} \end{cases}$$

(2) Population standard deviation: $\sigma = \sqrt{\sigma^2}$.

Example 2-3 (Variance)

Find the population variance and standard deviation for Example 2-2.

Solution. With $\mu = 1.9$,

$$\begin{aligned} \sigma^2 &= (0-1.9)^2 \tfrac{1}{10} + (1-1.9)^2 \tfrac{3}{10} + (2-1.9)^2 \tfrac{3}{10} + (3-1.9)^2 \tfrac{2}{10} + (4-1.9)^2 \tfrac{1}{10} \\ &= 1.29, \end{aligned}$$

and $\sigma = \sqrt{1.29} = 1.1358$.

2.2.3 Association: correlation

Table 2-5 gives two bivariate populations; each entry is the relative frequency of the pair (characteristic 1, characteristic 2).

Population A

char. 2 \ char. 1	0	1	2	Total
0	0.1	0.2	0.1	0.4
1	0.0	0.2	0.1	0.3
2	0.0	0.0	0.3	0.3
Total	0.1	0.4	0.5	1.0

Population B

char. 2 \ char. 1	0	1	2	Total
0	0.0	0.0	0.4	0.4
1	0.0	0.2	0.1	0.3
2	0.1	0.2	0.0	0.3
Total	0.1	0.4	0.5	1.0

The bottom margin gives the distribution of characteristic 1, the right margin that of characteristic 2. In both populations

$$\mu_1 = 1.4,\ \sigma_1 = 0.6633, \qquad \mu_2 = 0.9,\ \sigma_2 = 0.8307,$$

yet A's mass lies along a rising direction and B's along a falling one. Means and standard deviations do not describe a bivariate population; we need a measure of the relation between the characteristics. For a bivariate finite population $(c_{11}, c_{21}), \ldots, (c_{1N}, c_{2N})$ the **correlation coefficient** (Galton, Pearson) is

$$\rho = \frac{1}{N} \sum_{i=1}^{N} \left(\frac{c_{1i} - \mu_1}{\sigma_1} \right) \left(\frac{c_{2i} - \mu_2}{\sigma_2} \right).$$

Proposition (Range of the correlation coefficient)

$-1 \le \rho \le 1$, because

$$0 \le \frac{1}{N}\sum_{i=1}^{N}\left(\frac{c_{1i}-\mu_1}{\sigma_1} - \rho\,\frac{c_{2i}-\mu_2}{\sigma_2}\right)^2 = 1-\rho^2.$$

Proof. Write $u_i = (c_{1i}-\mu_1)/\sigma_1$ and $v_i = (c_{2i}-\mu_2)/\sigma_2$. By the definitions of σ_1^2, σ_2^2 and ρ,

$$\frac{1}{N}\sum u_i^2 = 1, \qquad \frac{1}{N}\sum v_i^2 = 1, \qquad \frac{1}{N}\sum u_i v_i = \rho.$$

Expanding the square,

$$\frac{1}{N}\sum(u_i-\rho v_i)^2 = \frac{1}{N}\sum u_i^2 - 2\rho\,\frac{1}{N}\sum u_i v_i + \rho^2\,\frac{1}{N}\sum v_i^2 = 1 - 2\rho^2 + \rho^2 = 1-\rho^2.$$

A mean of squares is ≥ 0, so $\rho^2 \le 1$. ■

Grouping distinct values c_{1i}^* $(i \le k)$ of characteristic 1 and c_{2j}^* $(j \le l)$ of characteristic 2, with f_{ij} the frequency of (c_{1i}^*, c_{2j}^*):

Definition 2-6 (Correlation of a finite population)

$$\rho = \sum_{i=1}^{k}\sum_{j=1}^{l}\left(\frac{c_{1i}^*-\mu_1}{\sigma_1}\right)\left(\frac{c_{2j}^*-\mu_2}{\sigma_2}\right)\frac{f_{ij}}{N} = \frac{\sum_{i=1}^{k}\sum_{j=1}^{l}(c_{1i}^*-\mu_1)(c_{2j}^*-\mu_2)\,f_{ij}/N}{\sigma_1\sigma_2}.$$

Example 2-4 (Correlations of Table 2-5)

Find the population correlation coefficients of populations A and B.
Solution. For A,

$$\begin{aligned}&\sum_i\sum_j (c_{1i}^*-\mu_1)(c_{2j}^*-\mu_2)\frac{f_{ij}}{N}\\ &\quad= (0-1.4)(0-0.9)(0.1) + (1-1.4)(0-0.9)(0.2) + \cdots + (2-1.4)(2-0.9)(0.3)\\ &\quad= 0.34,\end{aligned}$$

so $\rho = 0.34/(0.6633 \times 0.8307) = 0.6171$. The same computation for B gives a numerator of -0.46 and $\rho = -0.8348$.

For an infinite bivariate population let $p(x,y)$ be the limiting relative frequency of (x,y) (discrete case) or the joint density surface (continuous case): the **joint density function**. With μ_1, σ_1 and μ_2, σ_2 the means and standard deviations of the two characteristics:

Definition 2-7 (Correlation of an infinite population)

$$\rho = \begin{cases} \dfrac{\sum_{\text{all } x}\sum_{\text{all } y}(x-\mu_1)(y-\mu_2)\,p(x,y)}{\sigma_1\sigma_2} & \text{(discrete case)}\\[2ex] \dfrac{\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x-\mu_1)(y-\mu_2)\,p(x,y)\,dx\,dy}{\sigma_1\sigma_2} & \text{(continuous case)}\end{cases}$$

The numerator is the **covariance** of the two characteristics; the population covariance equals $\rho\sigma_1\sigma_2$. As in Example 2-4, $\rho > 0$ when the mass lies along a rising direction and $\rho < 0$ along a falling one; the more the mass concentrates on a line of nonzero slope, the closer ρ is to ± 1.

The population mean, variance, standard deviation, covariance and correlation—numbers describing a population—are called **parameters**. Section 2.3 shows how to picture the population from a sample; Section 2.4 gives sample quantities used to estimate parameters.

2.3 Graphical display of samples

Motivation. A sample is our only window on the population; tables and graphs of the sample give a first rough picture of the population's distribution.

2.3.1 Qualitative or few-valued data

A **frequency table** lists each value with its **frequency** (count) and **relative frequency** (count / sample size). It is drawn as a **bar graph** (bar heights proportional to relative frequencies) or a **pie graph** (sector areas, i.e. central angles, proportional to relative frequencies).

Example 2-5 (Bar and pie graphs)

Draw a bar graph and a pie graph of the sample in Table 1-1.
Solution. The bar graph has heights $0.42, 0.37, 0.15, 0.06$. The central angles of the pie are

$$360^\circ \times 0.42 = 151.2^\circ, \quad 360^\circ \times 0.37 = 133.2^\circ, \quad 360^\circ \times 0.15 = 54^\circ, \quad 360^\circ \times 0.06 = 21.6^\circ.$$

2.3.2 Frequency tables, histograms, polygons

For continuous values, or discrete values with many distinct values, divide the range into classes. Class widths are arbitrary, but each class should not have too small a relative frequency; too wide classes hide the shape; and no observation should fall on a class boundary.

Constructing a frequency table

(1) Range = (largest value) − (smallest value).
(2) Choose about 5–20 classes; class width = range ÷ (number of classes), rounded *up* to the smallest unit of the data.
(3) Start the first class at (smallest value) $-\frac{1}{2}\times$(smallest unit), and build the classes with the width from (2).
(4) Count the (relative) frequency of each class.

Example 2-6 (Heights of 40 people)

Heights (cm) of 40 randomly selected people:

183	168	165	158	180	176	168	160	160	150
180	183	170	170	185	170	165	170	168	154
152	173	173	165	163	163	160	178	138	171
163	168	165	165	178	170	168	183	145	174

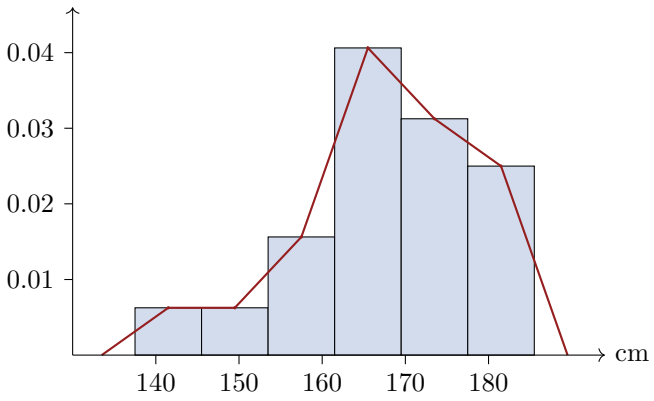
Make a frequency table.
Solution. Max 185, min 138, range 47. With 6 classes, $47/6 = 7.83\ldots$, rounded up to width 8. The first class starts at $138 - \frac{1}{2} \times 1 = 137.5$.

Class (cm)	Freq.	Rel. freq.
$[137.5, 145.5)$	2	0.050
$[145.5, 153.5)$	2	0.050
$[153.5, 161.5)$	5	0.125
$[161.5, 169.5)$	13	0.325
$[169.5, 177.5)$	10	0.250
$[177.5, 185.5)$	8	0.200
Total	40	1

A **histogram** marks the class intervals on the horizontal axis and draws over each class a rectangle with

$$\text{area} = \text{relative frequency}, \qquad \text{i.e.} \qquad \text{height} = \frac{\text{relative frequency}}{\text{class width}}.$$

Total area is then 1, so the histogram estimates the density curve. Classes need not have equal widths: narrow classes where data are dense show detail, wide classes where data are sparse collect enough values (Figure 2-13 splits the 4th class into two of width 4). A **frequency polygon** joins the midpoints of the tops of the rectangles, with end points on the axis half a class width beyond the first and last classes; its total area is also 1.



Histogram (height = rel. freq./8) and frequency polygon for Example 2-6 (Figures 2-12, 2-14)

2.3.3 Stem-and-leaf display

Write the leading digits (here the first two) in a column—the **stems**—with a vertical line to their right; write the last digit of each observation as a **leaf** to the right of its stem; finally sort the leaves. For Example 2-6:

18	0 0 3 3 3 5
17	0 0 0 0 0 1 3 3 4 6 8 8
16	0 0 0 3 3 3 5 5 5 5 5 8 8 8 8 8
15	0 2 4 8
14	5
13	8

This **stem-and-leaf display** looks like a histogram rotated by 90° (clockwise). The leaf width plays the role of the class width and must be constant; above it is 10, and splitting each stem in two (14^- for 140–145, 14^+ for 145–150) gives width 5. Its advantage is that the data values are kept; its drawback is that large samples are hard to draw.

2.3.4 Bivariate samples

The bivariate analogue of the frequency table is the **contingency table**, listing the (relative) frequency of each pair of values. For instance, the favourite subject of 400 students:

	Korean	English	Math	Total
Male	0.18	0.24	0.18	0.60
Female	0.14	0.16	0.10	0.40
Total	0.32	0.40	0.28	1.00

It can be drawn as a bivariate bar graph. For continuous (or many-valued) characteristics, class both ranges to get a contingency table, and draw a **bivariate histogram**: over each rectangular cell a box whose volume is the cell's relative frequency.

Example 2-7 (Height and weight)

Adding weight to the heights of Example 2-6 (height cm, weight kg):

(183, 82)	(168, 52)	(165, 52)	(158, 55)	(180, 93)
(176, 57)	(168, 60)	(160, 48)	(160, 50)	(150, 49)
(180, 79)	(183, 73)	(170, 60)	(170, 66)	(185, 74)
(170, 82)	(165, 57)	(170, 54)	(168, 57)	(154, 55)
(152, 50)	(173, 63)	(173, 60)	(165, 49)	(163, 63)
(163, 57)	(160, 57)	(178, 77)	(138, 54)	(171, 69)
(163, 50)	(168, 57)	(165, 54)	(165, 59)	(178, 79)
(170, 54)	(168, 79)	(183, 70)	(145, 37)	(174, 65)

Make a contingency table and a bivariate histogram.
Solution. Using the 6 height classes of Example 2-6 and 6 weight classes of width 10 starting at 36.5:

Height \ Weight	36.5–46.5	46.5–56.5	56.5–66.5	66.5–76.5	76.5–86.5	86.5–96.5	Total
137.5–145.5	0.025	0.025	0	0	0	0	0.050
145.5–153.5	0	0.050	0	0	0	0	0.050
153.5–161.5	0	0.100	0.025	0	0	0	0.125
161.5–169.5	0	0.125	0.175	0	0.025	0	0.325
169.5–177.5	0	0.050	0.150	0.025	0.025	0	0.250
177.5–185.5	0	0	0	0.075	0.100	0.025	0.200
Total	0.025	0.350	0.350	0.100	0.150	0.025	1.000

The bivariate histogram erects over each cell a box of volume equal to the entry (Figure 2-17).

2.3.5 Scatter plots and regressograms

A **scatter plot** plots each bivariate observation as the point (value 1, value 2). A **regressogram** classes one characteristic (as for a histogram) and, within each class, plots the mean of the other characteristic. For Example 2-7, classing weight into the 6 classes above, the mean heights are

Weight class	36.5–46.5	46.5–56.5	56.5–66.5	66.5–76.5	76.5–86.5	86.5–96.5
Mean height	145.0	159.9	168.3	180.5	176.2	180.0

Plotting these as a step function over the weight classes gives the regressogram (Figure 2-19). Both graphs show how the two characteristics vary together; the regressogram is generally more effective for analysing the relationship.

2.4 Summary measures of a sample

Motivation. Parameters are unknown; we estimate them by the corresponding quantities computed from the sample. Some care (trimming, dividing by $n-1$) makes these estimates better behaved.

2.4.1 Location

Let the sample values be $x_1, \dots, x_n$.

Definition 2-8 (Sample mean)

$$\bar{x} = \frac{1}{n}(x_1 + x_2 + \cdots + x_n) = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

The sample pth percentile is defined as for the population; the sample quartiles and median are $\hat{Q}_1, \hat{Q}_2, \hat{Q}_3$. They estimate the population mean and percentiles. A drawback of $\bar{x}$: a few values far from the bulk strongly affect it, and when the population spread is large some sample values may lie far on one side, making $\bar{x}$ a poor estimate. The $100p\%$ **trimmed mean** $\bar{x}_p$ removes the smallest and the largest $100p\%$ of the ordered values and averages the remaining $100(1-2p)\%$ (this p is unrelated to the p of percentiles).

Example 2-8 (Sample mean, quartiles, trimmed mean)

For the data of Example 2-6, find the sample mean, the sample quartiles and the 10% trimmed mean.

Solution. $\bar{x} = \frac{1}{40}(183 + 168 + \cdots + 174) = \frac{6698}{40} = 167.45$. From the stem-and-leaf display, $\hat{Q}_1$ is the midpoint of the 10th and 11th values, $\hat{Q}_2$ of the 20th and 21st, $\hat{Q}_3$ of the 30th and 31st:

$$\hat{Q}_1 = 163, \qquad \hat{Q}_2 = 168, \qquad \hat{Q}_3 = 173.5.$$

The 10% trimmed mean drops 4 values at each end ($138, 145, 150, 152$ and $183, 183, 183, 185$) and averages the middle 32:

$$\bar{x}_{0.1} = \frac{1}{32}(180 + 180 + \cdots + 154) = \frac{5379}{32} = 168.0938.$$

2.4.2 Spread

The **sample interquartile range** $\widehat{\mathrm{IQR}} = \hat{Q}_3 - \hat{Q}_1$ estimates the IQR. For the variance, the analogue of σ^2 would be $\frac{1}{n}\sum(x_i-\mu)^2$; replacing the unknown μ by $\bar{x}$ gives $\frac{1}{n}\sum(x_i-\bar{x})^2$, which tends to *underestimate* σ^2 because of the following identity.

Proposition (Centring at the sample mean)

For any number μ,

$$\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2 - (\bar{x}-\mu)^2.$$

Proof. [Supplement] Write $x_i - \mu = (x_i - \bar{x}) + (\bar{x} - \mu)$ and square:

$$\sum_{i=1}^{n}(x_i-\mu)^2 = \sum_{i=1}^{n}(x_i-\bar{x})^2 + 2(\bar{x}-\mu)\sum_{i=1}^{n}(x_i-\bar{x}) + n(\bar{x}-\mu)^2.$$

The middle sum is $\sum x_i - n\bar{x} = 0$. Divide by n and rearrange. ■

Hence dividing by n gives a value never larger than the "ideal" $\frac{1}{n}\sum(x_i-\mu)^2$. Dividing by $n-1$ instead slightly increases the estimate; this is known to be more accurate, and is the **sample variance**

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2.$$

Definition 2-9 (Sample standard deviation)

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2}.$$

Example 2-9 (Sample IQR and s.d.)

For the data of Example 2-6, find the sample interquartile range and the sample standard deviation.

Solution. From Example 2-8, $\widehat{\text{IQR}} = 173.5 - 163 = 10.5$. Next

$$s^2 = \frac{1}{40-1}\{(183-167.45)^2 + (168-167.45)^2 + \cdots + (174-167.45)^2\} = 108.6128,$$

so $s = \sqrt{108.6128} = 10.4218$.

2.4.3 Bivariate samples

For a bivariate sample $(x_1, y_1), \ldots, (x_n, y_n)$, the means and standard deviations of each characteristic are estimated separately:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n}x_i, \quad s_1 = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2}; \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n}y_i, \quad s_2 = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(y_i-\bar{y})^2}.$$

Definition 2-10 (Sample correlation coefficient)

$$r = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{x_i-\bar{x}}{s_1}\right)\left(\frac{y_i-\bar{y}}{s_2}\right) = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2}}.$$

(The two expressions agree since $(n-1)s_1s_2 = \sqrt{\sum(x_i-\bar{x})^2}\sqrt{\sum(y_i-\bar{y})^2}$.) The **sample covariance**, estimating the population covariance, is

$$rs_1s_2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y}).$$

Example 2-10 (Sample correlation)

Find the sample correlation coefficient for the data of Example 2-7.

Solution. For height, $\bar{x} = 167.45$, $s_1 = 10.42$ (Examples 2-8, 2-9); similarly for weight, $\bar{y} = 61.45$, $s_2 = 11.94$. The sample covariance is

$$rs_1s_2 = \frac{1}{40-1}\{(183-167.45)(82-61.45) + \cdots + (174-167.45)(65-61.45)\} = 90.59,$$

so $r = 90.59/(10.42 \times 11.94) = 0.73$.

Sample quantities such as the sample mean, variance, standard deviation, covariance and correlation are called **statistics**; a statistic used to estimate a parameter is an **estimator**.